Empirical Validation Protocol for D-TrOPD

Companion to "Decentralized Trust-Region Policy Distillation: A Theoretical Blueprint"

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Abstract

This protocol specifies the experiments required to validate the corrected D-TrOPD mechanism. Each experiment is tied to a specific theoretical claim in the companion theory paper, with a pre-registered hypothesis, the measurement that would confirm or refute it, and the ablation that isolates the responsible mechanism. Results from the earlier prototype are explicitly excluded: that implementation tested the superseded design (value-variance trust radius, reverse-/forward-KL direction switch, pooled-batch trust evaluation), all three of which the corrected theory identifies as flawed. This document is the skeleton of the empirical paper; sections marked [RESULTS] are to be filled after runs complete.

1 Implementation Checklist (Corrected Design)

Before any experiment counts as a test of D-TrOPD, the implementation must satisfy every item below. Each item corresponds to a documented failure mode of the prototype.

1. **Trust radius from self-nonstationarity, not value variance.** $\delta_i^{(t)} = \delta_{\text{base}} \exp(-\kappa \Delta_i^{(t)})$ with $\Delta_i^{(t)} = \bar{D}_{\text{KL}}(\pi_i^{(t)}, \pi_i^{(t-1)})$ on the agent's own rollout states. The value-variance radius must not appear anywhere in the code path.

2. **Single reverse-KL distillation with continuous attenuation.** No direction switch. Weight $w_{ij} = \exp(-\gamma[\text{softplus}(D_{ij} - \delta_i) - \text{softplus}(-\delta_i)])$, normalized so $w_{ij} = 1$ at $D_{ij} = 0$.

3. **Social divergence on own rollout states.** D_{ij} is evaluated under d_{π_i}, not on a pooled global batch. (The pooled batch is both a bound violation and a Byzantine spoofing surface.)

4. **Trained critic.** The value-regression loss $c_V \mathcal{L}_i^V$ is present in the optimized objective; verify by logging critic explained variance $\text{EV}_i > 0$ rising over training. A frozen or untrained critic invalidates both the advantages and the radius signal.

5. **Both trust regions enforced.** PPO clipping *and* KL early stopping at η_i; log the per-epoch update KL and confirm it saturates at η_i rather than exceeding it.

6. **Frozen teachers.** Neighbor snapshots detached and fixed for the epoch; verify no gradient flows to θ_j.

7. **Neighbor selection.** Default $\mathcal{N}_k(i) = \mathcal{N}_i$ (attenuation weights do the filtering). If a communication cap forces top-k, log the top-k set per epoch to quantify selection churn.

8. **Radius sensitivity scaled to the step cap.** Choose κ so $\kappa\eta_i \in [1, 3]$, giving the radius meaningful dynamic range.

2 Environment

Primary benchmark. Heterogeneous cooperative navigation in the Multi-Agent Particle Environment (MPE), 12 agents, 3 targets, with a capture rule requiring at least 3 agents at a target simultaneously (defeats lone-wolf strategies). Heterogeneity is enforced by partitioning the field:

- *High-friction zone*: baseline action velocity reduced by 50%.
- *Masked-observation zone*: sensor range reduced from $r = 1.5$ to $r = 0.5$.

Stressors (each toggled independently for ablations):

- Random motor failures locking an agent's movement mid-episode.
- Byzantine agents: a fraction f of agents broadcast adversarial snapshots (see H6).
- Communication dropout: 20% of inter-agent messages corrupted or dropped.
- Multi-modal goal stressor: symmetric target placement testing mode coverage.

Scale sweep. $N \in \{4, 8, 12, 24, 48\}$ with fixed interaction degree, for the size-dependence hypotheses (H4, H5).

3 Baselines

- **IPPO** ($\beta = 0$): isolates the value of any social term; also the degenerate limit of D-TrOPD.
- **MAPPO**: centralized critic; the "if you had a hub" upper reference.
- **HAPPO**: sequential updates; the "if you had a queue" guarantee reference.
- **Naive distillation** ($w_{ij} \equiv 1$): D-TrOPD with the trust machinery removed; must reproduce the distribution-mismatch collapse that motivates the method.
- **D-TrOPD (superseded design)**: value-variance radius + KL switch, as in the prototype; included once to demonstrate the documented pathologies on the record.

All methods share network sizes, GAE parameters, clipping ϵ, learning rate, and seeds. D-TrOPD vs. IPPO runs use identical seeds and environment instances for paired comparison.

4 Pre-Registered Hypotheses

Each hypothesis names the theoretical claim it tests, the measurement, and the decision criterion.

H1 (Prop. 2 of the theory paper: no chattering). The switched-objective baseline exhibits sign-flipping social gradients as edges cross the trust boundary; continuous attenuation does not. *Measure:* per-edge social-gradient direction autocorrelation and the count of sign flips per epoch, plotted against $|D_{ij} - \delta_i|$. *Confirmed if* the switched variant shows a spike in sign flips concentrated near the boundary that the attenuated variant lacks.

H2 (Prop. 3: forward-KL entropy inflation). Under large mismatch, the forward-KL fallback inflates student policy entropy proportionally to the squared mean gap; reverse-KL does not. *Measure:* student latent σ_i trajectories on edges with large D_{ij}, under each direction. *Confirmed if* σ_i^2 under forward-KL tracks $\sigma_j^2 + (\mu_i - \mu_j)^2$ while under reverse-KL it tracks σ_j^2.

H3 (Remark on the value-variance radius). The value-variance radius mis-ranks trust: a degenerate critic earns maximal radius; radii are not invariant to reward rescaling. *Measure:* radius δ_i under (a) an intentionally degenerate critic and (b) a $10\times$ reward rescale, for both radius definitions. *Confirmed if* the nonstationarity radius is invariant under (b) and contracts under instability, while the value-variance radius shows the predicted pathologies.

H4 (Thm. 1: simultaneity penalty scales with N). Without locality exploitation, the gap between the surrogate gain and realized improvement grows linearly in N at fixed η. *Measure:* per-round $L_{\pi^{(t)}}(\pi^{(t+1)}) - J(\pi^{(t+1)})$ across the scale sweep, densely coupled variant. *Confirmed if* the gap grows approximately linearly in N.

H5 (Thm. 2: size-independence under locality). With bounded interaction degree, the per-capita surrogate–return gap is flat in N. *Measure:* the same gap, per capita, across the scale sweep with fixed-degree topology; additionally measure the decay assumption directly by perturbing agent j's policy and recording the induced change in Q^i as a function of graph distance $\mathrm{dist}_G(i, j)$. *Confirmed if* the per-capita gap is approximately constant in N and the measured Q^i-sensitivity decays exponentially with hop distance (this validates Assumption 2 itself).

H6 (Prop. 5: Byzantine breakdown point). Performance degrades gracefully up to $f = 1/2$ adversarial trusted neighbors with robust aggregation, and collapses earlier without it. *Measure:* final return vs. Byzantine fraction $f \in \{0, 0.1, 0.2, 0.3, 0.4, 0.5\}$, with trimmed-mean vs. plain-sum aggregation, including an *adaptive* adversary that mimics honest gradients within the honest range (the worst case identified in the theory paper). *Confirmed if* trimmed-mean degrades smoothly to $f \approx 0.4$ while plain-sum collapses at small f; the adaptive adversary quantifies the residual bias.

H7 (headline): trust-gated distillation beats isolation under heterogeneity. D-TrOPD $>$ IPPO in sample efficiency and final return in the heterogeneous environment, and $\geq$ naive distillation everywhere (with naive distillation collapsing under mismatch). *Measure:* learning curves and final returns, 10 seeds, 95% bootstrap confidence intervals; report per-zone returns so improvement is not an artifact of one sector.

5 Ablations

Ablation	Variants	Isolates
Attenuation	continuous / hard switch / hard cutoff ($w \in \{0, 1\}$)	Prop. 2 mechanism
KL direction	reverse-only / forward-only / switched	Prop. 3 mechanism
Radius signal	nonstationarity / value-variance / fixed δ ($\kappa = 0$)	radius design
Trust-test batch	own rollout / pooled batch	Remark on pooled batches
Aggregation	trimmed mean / geometric median / plain sum	Prop. 5 mechanism
Neighbor selection	all neighbors / hard top-k / soft top-k	selection-boundary churn
β sweep	$\{0, 0.01, 0.05, 0.1, 0.5\}$, vs. $\beta_{\max}$ of Prop. 4	β-safety condition

6 Reporting Standards

- Minimum 10 seeds per configuration; report mean, 95% bootstrap CIs, and IQM (interquartile mean) following current MARL reporting practice.
- Paired seeds and identical environment instances for D-TrOPD vs. IPPO comparisons.
- Log throughout training: per-agent update KL vs. η_i; radius $\delta_i^{(t)}$; all D_{ij} and w_{ij}; critic explained variance; per-edge social-gradient norms.
- Negative results are reported. In particular, if H5's decay measurement fails (sensitivity does not decay with hop distance), Theorem 2's assumption is falsified for this environment and the paper must say so.
- Compute, wall-clock, and communication volume (bytes of snapshots exchanged) reported per method, since decentralization's cost–benefit claim depends on them.

7 Failure Modes to Watch (From the Prototype)

The earlier implementation produced poor results under the superseded design. Symptoms to monitor for regression:

- Sudden entropy blow-ups on individual agents (historically: forward-KL fallback under large mismatch — should be impossible now; if observed, the direction switch has leaked back in).
- Radii that grow as training destabilizes (historically: value-variance sign pathology).
- Oscillating inclusion/exclusion of the same neighbor across consecutive epochs (selection churn; mitigated by all-neighbor default).
- Trust scores that disagree between pooled-batch and own-rollout evaluation (spoofing surface; own-rollout is authoritative).

[Results] — to be completed

Sections for H1–H7 outcomes, ablation tables, and the Assumption-2 decay measurement, to be filled after the corrected implementation is run.